

Privacy Leakage and Lightweight Defenses in Graph Neural Networks for Sensitive Data

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Abstract

Membership inference attacks (MIAs) pose a significant privacy threat to machine learning models trained on sensitive data. While MIAs have been extensively studied for traditional neural networks, their behavior on graph neural networks (GNNs) remains less understood, particularly regarding the role of graph structure in amplifying or mitigating leakage. We present a systematic empirical study spanning 560 experiments across 8 datasets, 4 model architectures (including non-graph baselines), and 6 defense configurations. Our findings challenge prevailing assumptions: (1) GNNs exhibit lower membership leakage than non-graph models on citation networks, suggesting neighborhood aggregation acts as an implicit regularizer; (2) MIA vulnerability is primarily driven by model overfitting rather than graph structural properties; (3) label smoothing, commonly assumed to be privacy-protective, increases leakage by up to 8.5%; and (4) edge sparsification offers the most favorable privacy-utility tradeoff among lightweight defenses. These results provide actionable guidance for deploying GNNs on sensitive relational data.

1. Introduction

Graph neural networks (GNNs) have become the dominant paradigm for learning on relational data, with applications spanning social networks, biological interactions, financial transactions, and healthcare knowledge graphs [8, 9]. When deployed on sensitive data, these models inherit the privacy vulnerabilities of traditional machine learning, particularly susceptibility to membership inference attacks (MIAs) [6]. MIAs seek to determine whether a specific data point was used in a model's training set, posing serious risks to individual privacy, regulatory compliance (GDPR, HIPAA), and institutional trust.

Prior work on MIAs against GNNs has demonstrated that structural information can be a significant factor in privacy leakage [1, 2]. However, important questions remain unresolved: Does the graph structure inherently increase leakage compared to non-graph models? How do fundamental graph properties such as homophily and density modulate vulnerability? And which lightweight defenses offer practical protection without sacrificing model utility?

We address these questions through a comprehensive empirical study organized around four research questions:

RQ1: Do GNNs leak more membership information than non-graph baselines on identical data?

RQ2: How do graph homophily and edge density affect MIA vulnerability?

RQ3: Which lightweight defenses most effectively reduce membership leakage?

RQ4: What is the achievable privacy-utility Pareto frontier?

Our study encompasses 560 experiments across 8 datasets (2 real-world citation networks, 6 controlled synthetic graphs), 4 model architectures, and 6 defense configurations, each evaluated over 5 random seeds. Our key contributions are: (i) demonstrating that GNNs' neighborhood aggregation acts as an implicit

regularizer that reduces membership leakage relative to non-graph models; (ii) showing that MIA vulnerability is driven primarily by model overfitting, not graph structural properties per se; (iii) identifying that label smoothing counterintuitively increases leakage; and (iv) establishing that edge sparsification provides the best privacy-utility tradeoff among lightweight defenses.

2. Related Work

2.1 Membership Inference Attacks

Shokri et al. [6] introduced the shadow model paradigm for MIAs against machine learning classifiers, demonstrating that model confidence scores contain sufficient signal to distinguish training members from non-members. The key insight is that models tend to be more confident on training data than on unseen data, creating an exploitable signal. Subsequent work has extended these attacks to diverse architectures and domains, establishing MIAs as a fundamental privacy threat. Aerni et al. [5] recently highlighted that many privacy defense evaluations are misleading, as they fail to account for the baseline attack performance achievable without any model access, calling for more rigorous evaluation protocols.

2.2 MIAs on Graph Neural Networks

He et al. [1] formalized node-level membership inference against GNNs, showing that node features and graph topology jointly determine vulnerability. Their work introduced attack models specifically designed for the graph setting. Olatunji et al. [2] extended this analysis and demonstrated that structural information is a major contributing factor to leakage in GNNs, suggesting that the message-passing mechanism may amplify privacy risks. Yuan et al. [3] introduced the Generalized Homophily Ratio to characterize how local graph structure modulates privacy risk, providing a more nuanced understanding of graph-privacy interactions. Mueller et al. [4] explored privacy-utility trade-offs specifically in medical population graphs, finding that graph-based models do not

necessarily increase privacy risk, motivating our broader empirical investigation.

2.3 Privacy Defenses for GNNs

Defense strategies against MIAs span a spectrum from formal guarantees (differential privacy) to lightweight heuristics (regularization, output perturbation). Wu et al. [10] proposed graph perturbation as a defense mechanism specifically for GNNs, modifying the adjacency matrix to reduce information leakage. Rong et al. [7] introduced DropEdge as a regularization technique for improving GNN training depth, which has been repurposed as a potential privacy defense due to its stochastic edge removal during training. In this work, we systematically evaluate five lightweight defenses spanning training-time and inference-time interventions, focusing on approaches that require minimal computational overhead and are practical for real-world deployment.

3. Methodology

3.1 Datasets

We use two real-world citation networks and six controlled synthetic graphs designed to isolate the effects of graph properties. Cora (2,708 nodes, 5,429 edges, homophily $h=0.81$) and Citeseer (3,327 nodes, 4,732 edges, $h=0.74$) are standard benchmarks where nodes represent publications, edges represent citations, and node features are bag-of-words representations. For controlled analysis, we generate synthetic graphs using a stochastic block model with systematically varied homophily (high: $h=0.80$, low: $h=0.30$) and density (sparse: $d=0.005$, medium: $d=0.015$, dense: $d=0.040$), each with 1,000 nodes and well-separated Gaussian class features to enable near-perfect classification.

3.2 Model Architectures

We evaluate four architectures spanning non-graph and graph-based models: (i) Logistic Regression (LogReg), a linear baseline that ignores graph structure entirely; (ii) Multi-Layer Perceptron (MLP), a non-linear baseline that also ignores graph structure; (iii) Graph Convolutional Network (GCN) [8], using spectral-based neighborhood aggregation with symmetric normalization; and (iv) GraphSAGE [9], using sampling-based inductive aggregation with mean pooling. All neural models use 2 hidden layers with 64 units each, ReLU activations, dropout of 0.5, Adam optimizer ($\text{lr}=0.01$), and are trained for 200 epochs. We use a consistent 60/20/20 train/validation/test split across all experiments.

3.3 Attack Methodology

We implement two complementary MIA strategies following established protocols [1, 6]. The confidence-based attack trains a binary classifier (logistic regression) on shadow model confidence vectors to distinguish members from non-members. For each target model, we train a shadow model with the same architecture on disjoint data, then use its confidence outputs as labeled examples for the attack classifier. The threshold-based attack applies a calibrated threshold

directly to the target model's maximum prediction confidence, exploiting the observation that training members typically receive higher confidence scores. We report attack AUC as the primary metric, where 0.5 indicates no leakage and 1.0 complete leakage.

3.4 Defense Mechanisms

We evaluate five lightweight defenses chosen for practical deployability: (i) DropEdge [7]: randomly removes 50% of edges during each training epoch to reduce structural overfitting; (ii) Label Smoothing: replaces one-hot training labels with softened distributions using smoothing factor 0.1; (iii) Early Stopping: monitors validation loss and halts training when it begins to increase, preventing overfitting; (iv) Confidence Masking: at inference time, truncates output probability vectors to retain only the top-3 class probabilities, setting others to zero; and (v) Edge Sparsification: removes 20% of edges with lowest feature-similarity scores as a preprocessing step.

3.5 Experimental Protocol

Each configuration (dataset $\times$ model $\times$ defense) is evaluated across 5 random seeds (42, 123, 456, 789, 1024), yielding 560 total experiments. We report mean and standard deviation for all metrics. The attack evaluation uses balanced member/non-member sets drawn from the training and test splits, respectively. All experiments are conducted on a single GPU with PyTorch and PyTorch Geometric.

4. Results

4.1 RQ1: GNNs vs. Non-Graph Baselines

Table 1 presents our most surprising finding: on both citation networks, non-graph models exhibit substantially higher attack AUC than GNNs. On Cora, MLP achieves attack AUC of 0.769 compared to GCN's 0.554 and GraphSAGE's 0.635--a difference of 0.215 and 0.134 respectively. This pattern is even more pronounced on Citeseer, where MLP reaches 0.829 while GCN remains at 0.632. Notably, LogReg also shows high leakage (0.735 on Cora, 0.730 on Citeseer), indicating that the leakage difference is not merely a function of model complexity.

Table 1: Confidence-based attack AUC and test accuracy on citation networks (no defense). Std. dev. in parentheses.

Model	Cora AUC	Cora Acc	Cite. AUC	Cite. Acc
LogReg	0.735(.012)	0.743	0.730(.011)	0.707
MLP	0.769(.007)	0.727	0.829(.011)	0.711
GCN	0.554(.016)	0.872	0.632(.008)	0.724
GraphSAGE	0.635(.013)	0.880	0.720(.012)	0.749

This counterintuitive result can be explained by examining the generalization gap. Non-graph models, lacking access to structural information, achieve substantially lower test accuracy (MLP: 0.727 on Cora vs. GCN: 0.872). This larger train-test performance gap creates stronger signals for MIA exploitation, as the attack model can more easily distinguish the high-confidence training predictions from lower-confidence test predictions. GNNs' neighborhood aggregation

effectively regularizes predictions by smoothing them across connected nodes, reducing the per-node memorization that MIAs exploit. This finding challenges the assumption from prior work [2] that graph structure inherently increases privacy leakage, and instead suggests that the regularization effect of message passing may dominate (see Figure 1 for a visual comparison across all models).

4.2 RQ2: Homophily and Density Effects

On synthetic graphs with well-separated features, all models achieve near-perfect test accuracy (0.99-1.00), resulting in attack AUC near random chance (0.49-0.54). This confirms that MIA vulnerability is fundamentally driven by the generalization gap rather than graph structure per se: when models generalize perfectly, there is no train-test gap to exploit, regardless of graph properties.

The notable exception is GCN on low-homophily sparse graphs (AUC=0.592, accuracy=0.776), where the model struggles to leverage misaligned structural signals, creating a larger train-test gap. In contrast, GraphSAGE maintains near-perfect accuracy (1.000) and near-random attack AUC (0.523) on the same graphs, demonstrating its greater robustness to structural noise. This divergence highlights that privacy vulnerability is model-dependent even under identical graph conditions (see Figures 3 and 5 for detailed analysis).

4.3 RQ3: Defense Effectiveness

Table 2 presents defense effectiveness on Cora. The most notable and practically important finding is that label smoothing increases leakage for both GCN (+7.1%, from 0.554 to 0.593) and GraphSAGE (+8.5%, from 0.635 to 0.689). This occurs because label smoothing shifts the output confidence distribution toward a distinctive pattern--training examples still receive relatively higher confidence on their true class, but the smoothed distribution provides the attack model with a richer signal to exploit. This finding echoes Aerni et al.'s [5] warning that defense evaluations can be misleading.

Table 2: Defense effectiveness on Cora. Attack AUC (lower = better privacy) and test accuracy (higher = better utility).

Defense	GCN AUC	GCN Acc	SAGE AUC	SAGE Acc
None	0.554	0.872	0.635	0.880
DropEdge	0.559	0.874	0.667	0.875
Label Smooth.	0.593	0.873	0.689	0.878
Early Stop.	0.554	0.872	0.636	0.879
Conf. Masking	0.560	0.872	0.635	0.880
Edge Sparsif.	0.560	0.871	0.630	0.875

Edge sparsification provides the most consistent protection for GraphSAGE (AUC reduction from 0.635 to 0.630) with only marginal accuracy decrease (0.880 to 0.875). DropEdge, contrary to expectations, increases GraphSAGE vulnerability (0.635 to 0.667), likely because random edge removal during training introduces noise that widens the generalization gap rather than closing it. Early stopping and confidence masking show negligible effects, suggesting that the primary leakage

signal in GNNs is not concentrated in late-stage overfitting or tail confidence values (see Figure 4 for visual comparison).

4.4 RQ4: Privacy-Utility Frontier

Figure 2 maps the full privacy-utility landscape across all model-defense combinations. GCN achieves the most favorable operating point: attack AUC of 0.554 with test accuracy of 0.872, compared to GraphSAGE's AUC of 0.635 at accuracy 0.880. The marginal accuracy gain of GraphSAGE (+0.8 percentage points) comes at a substantial privacy cost (+14.6% relative leakage increase). Among defenses, edge sparsification and confidence masking define the Pareto frontier for GraphSAGE, while GCN's baseline already approaches the best achievable tradeoff. Non-graph baselines occupy the worst region of the frontier, with high leakage and low accuracy (see Figure 6 for a comprehensive heatmap across all configurations).

5. Discussion

5.1 The Regularization Hypothesis

Our central finding--that GNNs leak less than non-graph models--can be understood through the lens of implicit regularization. Neighborhood aggregation in GNNs forces each node's representation to be influenced by its neighbors, effectively smoothing the learned function across the graph. This smoothing reduces the model's capacity to memorize individual training examples, which is precisely the signal that MIAs exploit. The effect is analogous to how convolutional layers in CNNs share parameters across spatial locations, reducing memorization. This interpretation aligns with Mueller et al.'s [4] observation that graph structure does not necessarily increase privacy risk in medical settings, and extends it to a broader class of GNN architectures and datasets.

5.2 Why Label Smoothing Fails

The failure of label smoothing as a defense has important practical implications. Label smoothing is widely used as a regularizer in modern deep learning and has been suggested as a potential privacy protection mechanism. Our results show it can be counterproductive: by transforming the one-hot training signal into a distinctive softened distribution, it creates a confidence signature that is more informative to the attack model rather than less. Specifically, the smoothed confidence vectors for training members exhibit a characteristic pattern that differs systematically from test examples, providing the attack classifier with a stronger discriminative signal. This underscores the need for empirical validation of any defense mechanism [5].

5.3 Practical Recommendations

For practitioners deploying GNNs on sensitive relational data, our findings suggest the following guidelines: (1) Prefer GCN over GraphSAGE when the marginal accuracy gain does not justify the privacy cost--the 0.8% accuracy improvement may not warrant 14.6% higher leakage. (2) Apply edge sparsification as

a preprocessing step; it provides consistent, albeit modest, privacy improvement with minimal utility loss. (3) Avoid label smoothing as a privacy defense--it increases rather than decreases vulnerability. (4) Monitor the train-test accuracy gap as a proxy for privacy risk: larger gaps indicate greater MIA vulnerability regardless of model architecture.

5.4 Limitations and Future Work

Several limitations should be noted. First, we evaluate only confidence-based and threshold-based attacks; stronger attacks (e.g., loss-based, gradient-based, or adaptive attacks) may reveal different vulnerability patterns. Second, our synthetic graphs use well-separated Gaussian features, which may not reflect the complexity of real-world feature distributions. Third, we focus on transductive node classification and do not evaluate link-level or subgraph-level MIAs. Fourth, we do not evaluate differential privacy, which provides formal guarantees but at higher computational cost. Future work should extend this analysis to heterogeneous graphs, inductive settings, stronger adaptive attacks, and formal privacy mechanisms to provide a more complete picture of the GNN privacy landscape.

6. Conclusion

We present a systematic empirical study of membership inference vulnerability in graph neural networks, spanning 560 experiments across 8 datasets, 4 model architectures, and 6 defense configurations. Our results challenge prevailing assumptions about graph-based privacy risk: GNNs' neighborhood aggregation acts as an implicit regularizer that reduces membership leakage relative to non-graph baselines. We demonstrate that MIA vulnerability is fundamentally driven by model overfitting rather than graph structural properties, that label smoothing counterintuitively increases leakage by up to 8.5%, and that edge sparsification provides the best lightweight defense. These findings offer actionable guidance for the responsible deployment of GNNs on sensitive relational data and motivate further research into the interplay between graph structure, generalization, and privacy.

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